

exactly 80 mm: _____

FIRST SOLVE

New here? Do the walkthrough once at cubepath-six.vercel.app/ learn. This card is the memory jog.

NOTATION

R = L = turn face clockwise, looking at it from outside the cube.
 = counter-clockwise. 2 = half turn. Lowercase r f = that face plus the layer behind it, turning together.
 y = spin the whole cube left, white staying center. A whole-cube turn solves nothing.

WORDS

algorithm = a fixed sequence of turns + trigger = a short algorithm your hands run as one unit

1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm – work it out.

Can't see it? Build the four white edges on the YELLOW face first, each show its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on top, turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.



white sticker RIGHT
RUR'U



white sticker FRONT
L'LUL



white sticker UP run the first one once to knock it out, then look again
RUR'U

3 MIDDLE EDGES

Top edge with NO yellow: turn U until its front color matches the center under it. The arrow shows which slot it goes to.



goes RIGHT
U RUR'U y L'LUL



goes LEFT
U L'LUL y' RUR'U'

FIRST SOLVE

New! Here's the walk-through on cubeopath-3x3.vercel.app/learn. This card is the memory jog.

NOTATION

R = 1 turn that face clockwise, looking at it from outside the cube.
c = counter-clockwise. $2 = \text{half turn}$. Lowercase $r = f$ that face plus the layer behind it, turning together.
y = spin the whole cube left, leaving center. A whole-cube turn solves nothing.

WORDS

algorithm = a fixed sequence of turns • trigger = a short algorithm your hands run as one unit

1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm – work it out.

Can't see it? Build the four white edges on the YELLOW face first, each above its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on top, turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.



white sticker RIGHT
RUR'U



white sticker FRONT
L'ULU



white sticker U run the first one once to knock it out, then look again
RUR'U

3 MIDDLE EDGES

Top edge with NO yellow: Turn U until its front color matches the center inside it. The arrow shows which slot it goes to.

goes RIGHT

U **RUR'U** **y** **L'ULU**

goes LEFT

U **L'ULU** **y'** **RUR'U**

TWO-LOOK

OLL CROSS - yellow edges



Line bar left-to-right
FURU'U'F'



Hook hook pointing front-right
FURU'U'f'

OLL CORNERS - yellow face



Sune $\frac{1}{2}$ 1 yellow, rest clockwise
RUR'U' RU2R'



Anti-Sune $\frac{1}{2}$ 1 yellow, rest counter-clockwise
RU2R' U'RUR'



Pi $\frac{1}{2}$ 0 yellow: headlights left
FURU'U'f' FURU'U'F'

OLL CORNERS - continued



Headlights $\frac{1}{2}$ 2 yellow: headlights at the back
R2DR'U2 RD'R'U2R'



2x Headlights $\frac{1}{2}$ 0 yellow: headlights both sides
RUR'U' RU'R' URU2R'



Chameleon $\frac{1}{2}$ 2 yellow: next to each other
FUR'U'U'f' FRF'



Bowtie $\frac{3}{4}$ 2 yellow diagonal
F'URU'U'f' FR

FALLBACK

The badge on each row is your fallback: that many Sunes, with a U turn between, also finishes the case. Machine-checked: there is the word.

Top not all yellow and nothing matches? Run Sune and read again.

TWO-LOOK		OLL CORNERS - continued	
OLL CROSS - yellow edges			
	Line bar left-to-right FURU'U'F'		Headlights ≥ 2 yellow, headlights at the back R2DR'U2 RD'R'U2R'
	Hook hook pointing front-right fRU'R'U'f'		2x Headlights ≥ 6 yellow, headlights both sides RU'R'U RU'R' URU2R'
OLL CORNERS - yellow face			
	Sune ≥ 1 yellow, rest clockwise RU'R'U RU2R'		Chameleon ≥ 3 yellow, note to each other fUR'U'U'f' FRF'
	Anti-Sune ≥ 1 yellow, rest counter-clockwise RU2R' U'R'UR'		Bowtie ≥ 2 yellow diagonal F'fUR'U'U'f' FR
		FALLBACK	
		The badge on each row is your fallback: that many Sunes, with a U turn between, also finishes the case. Machine-checked; three is the worst. Top not all yellow and nothing matches? Run Sune and read again.	

ADJACENT CORNER SWAP

exactly one face shows headlight

T 1. headlights; pairs B-left F-left
RUR'U' R'F R2UR'U' RUR'F'

Ja 1. solid; pairs B-right F-left B-front
x R2FRF' RU2 r'Ur U2

Ib 1. solid; pairs B-left F-right B-back
RUR'F' RUR'U' R'F R2U'R'

Aa 1. headlights; pairs R-front
x L2D2 L'U'L D2 L'U'L'

Ab 1. headlights; pairs B-right R-back
x' L2D2 LUL' D2 LU'L

F 1. solid; pairs B-left F-right
R'U'F' RUR'U' R'F R2UR'U' RUR'UR'

One face shows headlights; two corners of that face match. Hold as headlight.

ADJACENT CORNER SWAP

continued

Ra 1. headlights; pairs B-left
RU'R'U' RUR DR'UR' R'U2R'

Rb 1. headlights; pairs B-left
R2F RURU'R' F'R U2R'U2R

Ga 1. headlights; pairs F-right
RUR'U'U' RU'R2 U'D R'UR'D'

Gb 1. headlights; pairs B-back
R'UR UD' R2UR'U'U' RU'R2D

Gc 1. headlights; pairs B-right
R2UR'URU'R R'UR2 UD' RU'R'D

Gd 1. headlights; pairs R-front
R'UR' UD' R2UR'U' UR'UR2D'

Learn in the printed order, three a week, about six weeks. Ja and Jb on the same flip — one is the mirror of the other.

ADJ-LOOK PLI

ADJACENT CORNER SWAP

exactly one face shows headlight

T 1. headlight: pairs B-left F-left
RUR'U' R'F R2UR'U' RUR'F'

Ja 1. solid: pairs B-right F-left B-front
x R2FRF' RU2 r'Ur U2

Jb 1. solid: pairs B-left F-right B-back
RUR'F' RUR'U' R'F R2UR'U'

Aa 1. headlight: pairs F-right F-front
x L2D2 L'U'L D2 L'U'L'

Ab 1. headlight: pairs B-right B-back
x' L2D2 LUL' L2 LU'L

F 1. solid:
RUR'F' RUR'U' R'F R2UR'U' RUR'UR'

One face shows headlight: two corners of that face match. Hold as headlight.

ADJACENT CORNER SWAP

continued

Ra 1. headlight: pairs B-left B-right
RU'R'U' RUR DR'UR'D' R'U2R'

Rb 1. headlight: pairs B-left B-front
R2F RURU'R' F'R U2R'U2R

Ga 1. headlight: pairs F-right F-front
RUR'U'U'U' RU'R2 U'D R'UR'D'

Gb 1. headlight: pairs B-back B-front
R'UR UD' R2UR'UR'U' RU'R2D

Gc 1. headlight: pairs B-right B-front
RU'R'UR'UR'U RU'R2 UD' RU'R'D'

Gd 1. headlight: pairs F-front F-right
RUR'U' UD' R2UR'U' UR'UR2D'

Learn in this group: three a week, about six weeks. Ja and Jb are the same flip — one is the mirror of the other.

ANXUS = BIG CUBES and THE MAP

BIG CUBES 4x4 and 5x5

Recall: build the 6 centers → pair the 12 edges → solve it as a 3x3. $3Rw = 2$ layers wide, $3Rw = 3$ layers, $2R = 2$ nd Layer ONLY → never widen it.

last 2 edges

4x4 PLL parity

4x4 PLL parity = 1 edge pair flipped - half the time

$Rw' U' R' U' R' F R F' Rw$ slice out, run it, slice back - both cubes

$2R2 U2 2R2 Uw2 2R2 Uw2$ 2 edge pairs swapped - half the time

$Rw' U2 x Rw U2 Rw U2 Rw'$ slice out, run it, slice back - half the time

$5x5$ edge parity = 1 edge pair flipped - half the time - *one more differs*; 5x5 has no PLL parity

$Rw U2 x Rw U2 Rw U2 3Rw U2 Lw U2 Rw' U2 Rw U2 Rw' U2 Rw'$

Recall: during the last layer, before OLL and PLL - both parity algorithms move corners.

NOTATION

Each letter turns that face clockwise, seen from outside the cube - so from your seat L and D and B look counter-clockwise. = counter-clockwise. 2 = half turn.

M = middle slice, turning the way L turns x y z = the whole cube, turning the way R, U and F turn. Lowercase f = that face plus the layer behind it, turning together.

BEGINNER SHORTCUT - use it until you know Sune

Twist a yellow corner four turns at a time, turning ONLY the top face between corners.

yellow faces RIGHT $R'D'RD x2$

yellow faces FRONT $D'R'DR x2$

ANXEN – RIG CUBES AND THE MAP
RIG CUBES 4x4 and 5x5
 Reduce, making 6 centers → pair the 12 edges → solve it as a 3x3. Rw = 2 layers wide, 3Rw = 3 layers, 2R = 2nd LAYER only – never widen it.
 last 2 edges
 4x4 PLL parity
 $\begin{matrix} \text{Rw'} & \text{U' R' U} & \text{F' R F' R'} & \text{Rw} \\ 2\text{R}2\text{U} & 2\text{R}2\text{U} & \text{Uw}2 & 2\text{R}2\text{U} & \text{Uw}2 \end{matrix}$ slice out, run it, slice back – both cubes
 4x4 OLL parity = 1 edge pair flipped - half the time
 2 edge pairs swapped - half the time
 $\begin{matrix} \text{Rw} & \text{U}2 & \text{x} & \text{Rw} & \text{U}2 & \text{Rw}' & \text{U}2 & \text{Rw}' \\ \text{5x5 edge parity} & = & 1 \text{ edge pair flipped - half the time - one more differs; 5x5 has no PLL parity} \end{matrix}$
 $\begin{matrix} \text{Rw} & \text{U}2 & \text{x} & \text{Rw} & \text{U}2 & \text{3Rw} & \text{U}2 & \text{Lw} & \text{U}2 & \text{Rw}' & \text{U}2 & \text{Rw} & \text{U}2 & \text{Rw}' & \text{U}2 & \text{Rw}' \end{matrix}$
 Notation during the last layer, before OLL and PLL – both parity algorithms move corners
NOTATION
 Each letter turns that face clockwise,
 seen from outside the cube – so from your seat! D and B look counter-clockwise = counter-clockwise. 2 = half turn.
 M = middle slice, turning the way L turns. x y z = the whole cube, turning the way R, U and F turn. Lowercase f = that face plus the layer behind it, turning together.

WORDS
 algorithm = a fixed sequence of turns - trigger = a short algorithm you hands-run one unit - glued move = the R U F U trigger - sledalgorithm = a case like R F R F trigger - parity = a case a 3c3 cannot have - edge pair = two edges - shy move = on a 4x4 or 5x5.

exactly 90 mm; 1

Print at 100% / Actual size — do NOT fit to page. Two-sided, flip on the LONG edge. The numeral on both faces is your check: cut ONE card first. This bar must measure exactly 80 mm: 

FIRST SOLVE 4 YELLOW CROSS - yellow edges only — ignore the corners  Dot no yellow edge at all FRUR'U'F'  Hook two at a right angle: hold them BACK and LEFT FRUR'U'F'  Line hold the bar left-to-right FRUR'U'F' One algorithm, up to three times: dot to hook to line to cross. 5 MATCH THE YELLOW EDGES First turn U so at least two top edges match the center under them. This algorithm is called <i>Sune</i>; you use it on every card after this one.  two matched, side by side hold them BACK and LEFT RUR'U RU2R'  two matched, opposite run it once from any hold, then look again RUR'U RU2R' LAST-LAYER ORDER ON THIS CARD: yellow cross, match edges, place corners, twist corners. Card 2 replaces this order permanently — it is the only thing on this card that gets retired. UNLOCK CARD 2 — five scrambles in a row with this card face down. ◻ MASTER: those five under 3:00. My target: _____. -NEXT: 2/3 TWO-LOOK 	6 PUT THE YELLOW CORNERS HOME  1/3 - B  one corner home hold it FRONT-LEFT RU'L'U RU'L  no corner home run it once from any hold, then look again RU'L'U RU'L This cycles the other three corners into place. It also twists the yellow face you just built — expected, the next step rebuilds it. 7 TWIST THE YELLOW CORNERS  yellow faces RIGHT R'D'RD x2  yellow faces FRONT D'R'DR x2 Between corners turn ONLY the top face, to bring the next unsolved corner to front-right. Never turn the whole cube. The cube will look destroyed while you do this. Keep going — the last top turn brings it back. LAST-LAYER ORDER ON THIS CARD: yellow cross, match edges, place corners, twist corners. Card 2 replaces this order permanently — it is the only thing on this card that gets retired. UNLOCK CARD 2 — five scrambles in a row with this card face down. ◻ MASTER: those five under 3:00. My target: _____. -NEXT: 2/3 TWO-LOOK 	FIRST SOLVE 4 YELLOW CROSS - yellow edges only — ignore the corners  Dot no yellow edge at all FRUR'U'F'  Hook two at a right angle: hold them BACK and LEFT FRUR'U'F'  Line hold the bar left-to-right FRUR'U'F' One algorithm, up to three times: dot to hook to line to cross. 5 MATCH THE YELLOW EDGES First turn U so at least two top edges match the center under them. This algorithm is called <i>Sune</i>; you use it on every card after this one.  two matched, side by side hold them BACK and LEFT RUR'U RU2R'  two matched, opposite run it once from any hold, then look again RUR'U RU2R' LAST-LAYER ORDER ON THIS CARD: yellow cross, match edges, place corners, twist corners. Card 2 replaces this order permanently — it is the only thing on this card that gets retired. UNLOCK CARD 2 — five scrambles in a row with this card face down. ◻ MASTER: those five under 3:00. My target: _____. -NEXT: 2/3 TWO-LOOK 	6 PUT THE YELLOW CORNERS HOME  1/3 - B  one corner home hold it FRONT-LEFT RU'L'U RU'L  no corner home run it once from any hold, then look again RU'L'U RU'L This cycles the other three corners into place. It also twists the yellow face you just built — expected, the next step rebuilds it. 7 TWIST THE YELLOW CORNERS  yellow faces RIGHT R'D'RD x2  yellow faces FRONT D'R'DR x2 Between corners turn ONLY the top face, to bring the next unsolved corner to front-right. Never turn the whole cube. The cube will look destroyed while you do this. Keep going — the last top turn brings it back. LAST-LAYER ORDER ON THIS CARD: yellow cross, match edges, place corners, twist corners. Card 2 replaces this order permanently — it is the only thing on this card that gets retired. UNLOCK CARD 2 — five scrambles in a row with this card face down. ◻ MASTER: those five under 3:00. My target: _____. -NEXT: 2/3 TWO-LOOK 
TWO-LOOK PLL CORNERS - headlights = two matching corners on one face  T headlights on ONE face — hold that face LEFT RUR'U R'F R2UR'RU' RUR'F'  Y no headlights — the two swapping corners sit diagonally FRUR'U' RUR'F' RUR'U' R'FRF' PLL EDGES - corners home, top still not solved  Ub the front edge travels LEFT R2U RUR'U RU' R'UR'  Ua the front edge travels RIGHT M2U MU2 M'UM2  H both pairs swap straight across M2UM2 U2 M2UM2  Z each pair swaps with its neighbour M'U' M2UM2U' MU2 M2U THE NEW ORDER: yellow cross, yellow face, corners home, edges home. This never changes again. Card 1's order is retired, and so is Niklas; it moves corners but wrecks the yellow face, and here the yellow face is finished first. UNLOCK CARD 1 — fifteen last layers in a row, each finished in exactly two algorithms, case named out loud first. A repeat does not count. ◻ MASTER: solves averaging under 1:00. My target: _____. -NEXT: 3/3 ONE-LOOK PLL 	START HERE - two new algorithms finish any last layer Yellow cross: F R U R' U' F until the cross appears — at most 3. Yellow face: <i>Sune</i>, read again, repeat — at most 3 (see the badges). Corners: T-Perm. No headlights to hold left? Run it twice. Edges: Ub. No solved edge to put at the back? Run it twice. Each case on the front replaces one repeat with one run. Three a week; you can solve the whole time. HVS Z Both show headlights on all four faces. The edge between them decides: it belongs to the opposite face = H, to a neighbour = Z. STUCK? Corners home but nothing solves? Turn the top face once and read again — that is often the whole fix. One case has beaten you five times? Park it, use its fallback above, come back next week. A single piece looks twisted and no algorithm touches it? The cube was reassembled wrong. Pop it out and reset it. WORDS OLL = make the whole top face yellow - PLL = slide the top pieces home - headlights = two matching corners on one face - AUF = the free top turn that lines a case up LAST-LAYER ORDER ON THIS CARD: yellow cross, match edges, place corners, twist corners. Card 2 replaces this order permanently — it is the only thing on this card that gets retired. UNLOCK CARD 2 — five scrambles in a row with this card face down. ◻ MASTER: those five under 3:00. My target: _____. -NEXT: 2/3 TWO-LOOK 	TWO-LOOK PLL CORNERS - headlights = two matching corners on one face  T headlights on ONE face — hold that face LEFT RUR'U R'F R2UR'RU' RUR'F'  Y no headlights — the two swapping corners sit diagonally FRUR'U' RUR'F' RUR'U' R'FRF' PLL EDGES - corners home, top still not solved  Ub the front edge travels LEFT R2U RUR'U RU' R'UR'  Ua the front edge travels RIGHT M2U MU2 MUM2  H both pairs swap straight across M2UM2 U2 M2UM2  Z each pair swaps with its neighbour M'U' M2UM2U' MU2 M2U THE NEW ORDER: yellow cross, yellow face, corners home, edges home. This never changes again. Card 1's order is retired, and so is Niklas; it moves corners but wrecks the yellow face, and here the yellow face is finished first. UNLOCK CARD 1 — fifteen last layers in a row, each finished in exactly two algorithms, case named out loud first. A repeat does not count. ◻ MASTER: solves averaging under 1:00. My target: _____. -NEXT: 3/3 ONE-LOOK PLL 	START HERE - two new algorithms finish any last layer Yellow cross: F R U R' U' F until the cross appears — at most 3. Yellow face: <i>Sune</i>, read again, repeat — at most 3 (see the badges). Corners: T-Perm. No headlights to hold left? Run it twice. Edges: Ub. No solved edge to put at the back? Run it twice. Each case on the front replaces one repeat with one run. Three a week; you can solve the whole time. HVS Z Both show headlights on all four faces. The edge between them decides: it belongs to the opposite face = H, to a neighbour = Z. STUCK? Corners home but nothing solves? Turn the top face once and read again — that is often the whole fix. One case has beaten you five times? Park it, use its fallback above, come back next week. A single piece looks twisted and no algorithm touches it? The cube was reassembled wrong. Pop it out and reset it. WORDS OLL = make the whole top face yellow - PLL = slide the top pieces home - headlights = two matching corners on one face - AUF = the free top turn that lines a case up LAST-LAYER ORDER ON THIS CARD: yellow cross, match edges, place corners, twist corners. Card 2 replaces this order permanently — it is the only thing on this card that gets retired. UNLOCK CARD 2 — five scrambles in a row with this card face down. ◻ MASTER: those five under 3:00. My target: _____. -NEXT: 2/3 TWO-LOOK 
ONE-LOOK PLL EDGES ONLY - corners already home  • Ua B solid, F L R headlights; 3 edges counter-clockwise M2U MU2 M'UM2  • Ub B solid, F L R headlights; 3 edges clockwise R2U RUR'U RU' R'UR'  • H all 4 headlights; opposite edges swap M2UM2 U2 M2UM2  • Z all 4 headlights; adjacent edges swap M'U' M2UM2U' MU2 M2U HOW TO LOOK 1. Count the headlights faces. One = front of this card. Four = this column. None = the column on the right. 2. Only then read the edges to pick the exact case. 3. Line the case up as drawn, run it, turn the top again to finish — that last free turn is the AUF. • = already yours from Card 2. WORDS PLL = slide the top pieces home - headlights = two matching corners on one face - AUF = the free top turn that lines a case up - adjacent corner swap = the swapping corners share an edge - diagonal corner swap = the swapping corners sit opposite Card 2 gives you 19/72 last layers in one look. This card gives 71/72. Next: the other 57 top-face cases, and the first two layers solved as pairs — drilled with a cube in the app: cubepath-six-vercel.app/practice DONE — all twenty-one named correctly, twenty-one in a row, no card in hand, on two separate days. ◻ MASTER: PLL under 5 s, solves under 0:30. My target: _____. 	DIAGONAL CORNER SWAP - no headlights anywhere  • Y pairs F-left B-back FRUR'U' RUR'F' RUR'U' R'FRF'  • Y pairs F-left L-front R'UR'U y R'F' R2UR'U R'FRF'  • Na pairs B-left F-right L-front B-back RUR'U RUR'F' RUR'U' R'F R2UR' U2UR'R'  • Nb pairs B-right F-left L-back B-front R' UR'U'R' F'UF' RUR' FR'F' RU'R'  • E nothing matches X' L'ULD' L'U'LD L'U'LD' L'ULD' STUCK? Case you have not learned? Two-look it from Card 2 — T-Perm then Ub still solves it. Nothing here can strand you. Beaten five times by one case? Park it, two-look it, come back next week. LAST-LAYER ORDER ON THIS CARD: yellow cross, match edges, place corners, twist corners. Card 2 replaces this order permanently — it is the only thing on this card that gets retired. UNLOCK CARD 2 — five scrambles in a row with this card face down. ◻ MASTER: PLL under 5 s, solves under 0:30. My target: _____. 	ONE-LOOK PLL EDGES ONLY - corners already home  • Ua B solid, F L R headlights; 3 edges counter-clockwise M2U MU2 M'UM2  • Ub B solid, F L R headlights; 3 edges clockwise R2U RUR'U RU' R'UR'  • H all 4 headlights; opposite edges swap M2UM2 U2 M2UM2  • Z all 4 headlights; adjacent edges swap M'U' M2UM2U' MU2 M2U HOW TO LOOK 1. Count the headlights faces. One = front of this card. Four = this column. None = the column on the right. 2. Only then read the edges to pick the exact case. 3. Line the case up as drawn, run it, turn the top again to finish — that last free turn is the AUF. • = already yours from Card 2. WORDS PLL = slide the top pieces home - headlights = two matching corners on one face - AUF = the free top turn that lines a case up - adjacent corner swap = the swapping corners share an edge - diagonal corner swap = the swapping corners sit opposite Card 2 gives you 19/72 last layers in one look. This card gives 71/72. Next: the other 57 top-face cases, and the first two layers solved as pairs — drilled with a cube in the app: cubepath-six-vercel.app/practice DONE — all twenty-one named correctly, twenty-one in a row, no card in hand, on two separate days. ◻ MASTER: PLL under 5 s, solves under 0:30. My target: _____. 	DIAGONAL CORNER SWAP - no headlights anywhere  • Y pairs F-left B-back FRUR'U' RUR'F' RUR'U' R'FRF'  • Y pairs F-left L-front R'UR'U y R'F' R2UR'U R'FRF'  • Na pairs B-left F-right L-front B-back RUR'U RUR'F' RUR'U' R'F R2UR' U2UR'R'  • Nb pairs B-right F-left L-back B-front R' UR'U'R' F'UF' RUR' FR'F' RU'R'  • E nothing matches X' L'ULD' L'U'LD L'U'LD' L'ULD' STUCK? Case you have not learned? Two-look it from Card 2 — T-Perm then Ub still solves it. Nothing here can strand you. Beaten five times by one case? Park it, two-look it, come back next week. LAST-LAYER ORDER ON THIS CARD: yellow cross, match edges, place corners, twist corners. Card 2 replaces this order permanently — it is the only thing on this card that gets retired. UNLOCK CARD 2 — five scrambles in a row with this card face down. ◻ MASTER: PLL under 5 s, solves under 0:30. My target: _____. 
ANNEX — BIG CUBES AND THE MAP  A - B THE LADDER 1/3 FIRST SOLVE - 9 algorithms - unlock: five solves, card face down. 2/3 TWO-LOOK - +13 - unlock: fifteen last layers in exactly two algorithms. 3/3 ONE-LOOK PLL - +15 - done: all 21 named, twice on separate days. After that: full OLL, F2L, cross planning, look-ahead — in the app, not on card stock. WHY THERE ARE ONLY THREE A card is good at a closed set of cases you tell apart by sight. It is bad at a skill you drill. F2L, cross planning and look-ahead are drills with no finite case list. Full OLL is 57 cases, 22 of which differ only in a slider a fifth of a millimetre wide at this size. All of them are in the app, with a real cube and randomised setup. This set stops where the medium stops. PRINT AND VERSION Every ALGORITHM on these cards is expanded from machine-verified data and checked against a cube simulator at build time; if one were wrong the build would fail rather than slip. Recognition wording is generated from the same permutation the diagram is drawn from. Set v1.0, reprint any single card at cubepath-six-vercel.app/print - print at 100% / Actual size, never - Fit to page. Not a tier, no unlock, no order. Use it when you buy a 4x4. 	WORDS OLL = make the whole top face yellow - PLL = slide the top pieces home - <i>Sune</i> = the one-yellow-corner algorithm, used on every card after Card 1 - sexy move = the R U R' U' trigger - sledgelineamer = the R' F R F' trigger - headlights = two matching corners on one face - trigger = a short algorithm your hands run as one unit - AUF = the free top turn that lines a case up parity = a case a 3x3 cannot have - edge pair = two edges glued into one, on a 4x4 or 5x5 - F2L = first two layers, solved as pairs - look-ahead = watching the next piece while your hands finish this one - adjacent corner swap = the swapping corners share an edge - diagonal corner swap = the swapping corners sit opposite LAST-LAYER ORDER ON THIS CARD: yellow cross, match edges, place corners, twist corners. Card 2 replaces this order permanently — it is the only thing on this card that gets retired. UNLOCK CARD 2 — five scrambles in a row with this card face down. ◻ MASTER: PLL under 5 s, solves under 0:30. My target: _____. 	ANNEX — BIG CUBES AND THE MAP  A - B THE LADDER 1/3 FIRST SOLVE - 9 algorithms - unlock: five solves, card face down. 2/3 TWO-LOOK - +13 - unlock: fifteen last layers in exactly two algorithms. 3/3 ONE-LOOK PLL - +15 - done: all 21 named, twice on separate days. After that: full OLL, F2L, cross planning, look-ahead — in the app, not on card stock. WHY THERE ARE ONLY THREE A card is good at a closed set of cases you tell apart by sight. It is bad at a skill you drill. F2L, cross planning and look-ahead are drills with no finite case list. Full OLL is 57 cases, 22 of which differ only in a slider a fifth of a millimetre wide at this size. All of them are in the app, with a real cube and randomised setup. This set stops where the medium stops. PRINT AND VERSION Every ALGORITHM on these cards is expanded from machine-verified data and checked against a cube simulator at build time; if one were wrong the build would fail rather than slip. Recognition wording is generated from the same permutation the diagram is drawn from. Set v1.0, reprint any single card at cubepath-six-vercel.app/print - print at 100% / Actual size, never - Fit to page. Not a tier, no unlock, no order. Use it when you buy a 4x4. 	WORDS OLL = make the whole top face yellow - PLL = slide the top pieces home - <i>Sune</i> = the one-yellow-corner algorithm, used on every card after Card 1 - sexy move = the R U R' U' trigger - sledgelineamer = the R' F R F' trigger - headlights = two matching corners on one face - trigger = a short algorithm your hands run as one unit - AUF = the free top turn that lines a case up parity = a case a 3x3 cannot have - edge pair = two edges glued into one, on a 4x4 or 5x5 - F2L = first two layers, solved as pairs - look-ahead = watching the next piece while your hands finish this one - adjacent corner swap = the swapping corners share an edge - diagonal corner swap = the swapping corners sit opposite LAST-LAYER ORDER ON THIS CARD: yellow cross, match edges, place corners, twist corners. Card 2 replaces this order permanently — it is the only thing on this card that gets retired. UNLOCK CARD 2 — five scrambles in a row with this card face down. ◻ MASTER: PLL under 5 s, solves under 0:30. My target: _____. 

Print at 100% / Actual size — do NOT fit to page. Two-sided, flip on the LONG edge. The numeral on both faces is your check: cut ONE card first. This bar must measure exactly 80 mm: 